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SECTION – A

Question numbers 1 to 6 carry 1 mark each.

1. If vectors $\vec{a}$ and $\vec{b}$ are such that $|\vec{a}| = \frac{1}{2}$, $|\vec{b}| = \frac{4}{\sqrt{3}}$ and $|\vec{a} \times \vec{b}| = \frac{1}{\sqrt{3}}$, then find $|\vec{a} \cdot \vec{b}|$.

2. If $\vec{a}$ and $\vec{b}$ are unit vectors, then what is the angle between $\vec{a}$ and $\vec{b}$ for $\vec{a} - \sqrt{2}\vec{b}$ to be a unit vector?

3. Find the distance between the planes $\vec{r} \cdot (2\hat{i} - 3\hat{j} + 6\hat{k}) - 4 = 0$ and $\vec{r} \cdot (6\hat{i} - 9\hat{j} + 18\hat{k}) + 30 = 0$.

4. If A is a square matrix such that $|A| = 5$, write the value of $|AA^T|$.

5. If $A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -4 \\ 3 & -2 \end{bmatrix}$, find $|AB|$.

6. If $A = \begin{bmatrix} 0 & 3 \\ 2 & -5 \end{bmatrix}$ and $kA = \begin{bmatrix} 0 & 4a \\ -8 & 5b \end{bmatrix}$, find the values of k and a.

SECTION – B

Question numbers 7 to 19 carry 4 marks each.

7. Differentiate $(\sin 2x)^x + \sin^{-1}\sqrt{3x}$ with respect to x.

OR

Differentiate $\tan^{-1}\left(\dfrac{\sqrt{1+x^2} - \sqrt{1-x^2}}{\sqrt{1+x^2} + \sqrt{1-x^2}}\right)$ with respect to $\cos^{-1} x$.

8. Find k, if $f(x) = \begin{cases} k \sin \frac{\pi}{2}(x+1), & x \le 0 \\ \frac{\tan x - \sin x}{x^3}, & x > 0 \end{cases}$ is continuous at $x = 0$.

9. Find equation of normal to the curve $ay^2 = x^3$ at the point whose x coordinate is am^2.

10. Find: $\displaystyle\int \frac{1 - \sin x}{\sin x(1 + \sin x)}\, dx$

11. Find: $\displaystyle\int \left[\log(\log x) + \frac{1}{(\log x)^2}\right] dx$

12. Evaluate: $\displaystyle\int_0^{\pi/2} \frac{\sin^2 x}{\sin x + \cos x}\, dx$

OR

Evaluate: $\displaystyle\int_0^1 \cot^{-1}(1 - x + x^2)\, dx$

13. Solve the differential equation: $(x+1)\dfrac{dy}{dx} - y = e^3 x(x+1)^3$